

HACKMYSURU 1.0 — CIVICPULSE

1-PAGE DECISION LOG | Civic Complaint Follow-through

PROBLEM CHOSEN	Sub-problem #2 — Follow-through. Citizens can report civic issues, but the real challenge is what happens after reporting: complaints may remain open, receive little action, or lose visibility over time. CivicPulse focuses specifically on keeping the complaint journey visible and measurable.
CORE SOLUTION	CivicPulse is a citizen-facing web application where a citizen submits a complaint, receives a unique complaint ID, sees its current status and assigned department information, and can track the complaint over time. The system is designed to estimate which complaints are at risk of being forgotten so attention can be focused before follow-through is lost.
KEY DESIGN DECISIONS	• Deep focus on follow-through rather than building a full municipal platform. • Citizen-first interface: simple reporting and complaint tracking. • Unique complaint ID for every submission. • Timeline-based tracking to make complaint progress understandable. • Explainable rule-based risk scoring for the MVP instead of an opaque ML model. • No routing/GIS system, department dashboard, chatbot, mobile app, image detection, or warning/notification system in the MVP scope.
RISK INTELLIGENCE DECISION	The MVP risk model estimates follow-through risk on a 0–100 scale: LOW (0–39), MEDIUM (40–69), HIGH (70–100). Factors considered include complaint age, time since the last action/update, issue type, historical delay patterns, acknowledgement/update status, and similar unresolved complaints. Recent action or resolution reduces estimated risk. The score is an explainable MVP estimate, not a scientifically validated probability.
TECHNICAL APPROACH	Frontend: HTML/CSS with a responsive citizen-facing interface. Backend: Python Flask. Complaint records are handled by the Flask application and exposed through dedicated submission and tracking flows. The MVP is deployed as a public web service on Render and maintained in a public GitHub repository.
DATA & RESPONSIBLE ASSUMPTIONS	No fabricated MCC operational data is presented as real. Any historical patterns used for demonstration are synthetic/demo assumptions unless verified from an official source. The system is designed so real historical complaint data can later be used to validate and train a stronger predictive model.
TRADE-OFFS & FUTURE WORK	We chose a small, explainable MVP that can be demonstrated reliably within the hackathon time limit rather than attempting an end-to-end civic platform. Future versions can add persistent production-grade storage, validated historical data, trained ML models, stronger status/action histories, and citizen notifications/escalation after the core follow-through workflow is proven.